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PROBLEM & USERS: ECOSYSTEMS ACROSS AN AREA

How could new development affect an area's water supply and rivers through the year?

PROJECT GOAL

Water available each season
Water used and returned
Town reserves + local rivers

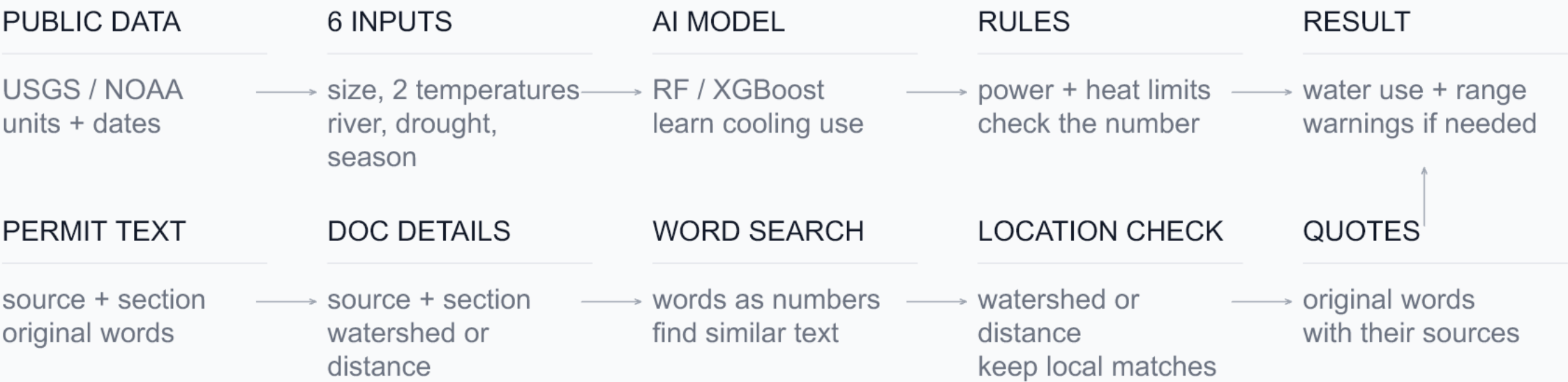
WORKING DEMO

Weather data + cooling-water model
Rules check numbers + permit quotes
Full area assessment still to build

Cooling is one example. Users: town planners, researchers, water providers.

SEMANTIC MAP: CURRENT SYSTEM

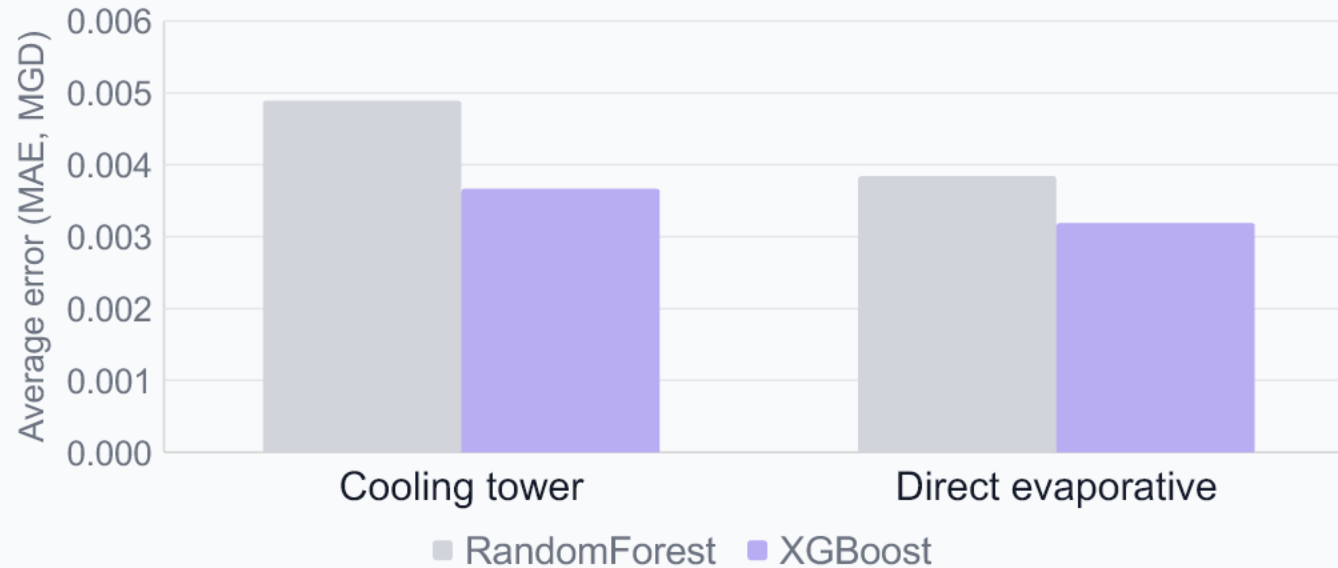
Demo request: equipment size, cooling type, watershed code, month. Exact location is optional.



The AI estimates water use. Rules check it. Local permit quotes explain the context.

E3: forced wrong results stay within 0-0.446484 MGD at 40 MW. E4: all 4 location tests pass.

EXPERIMENT E1: COMPARING AI MODELS



XGBOOST: LOWER ERROR

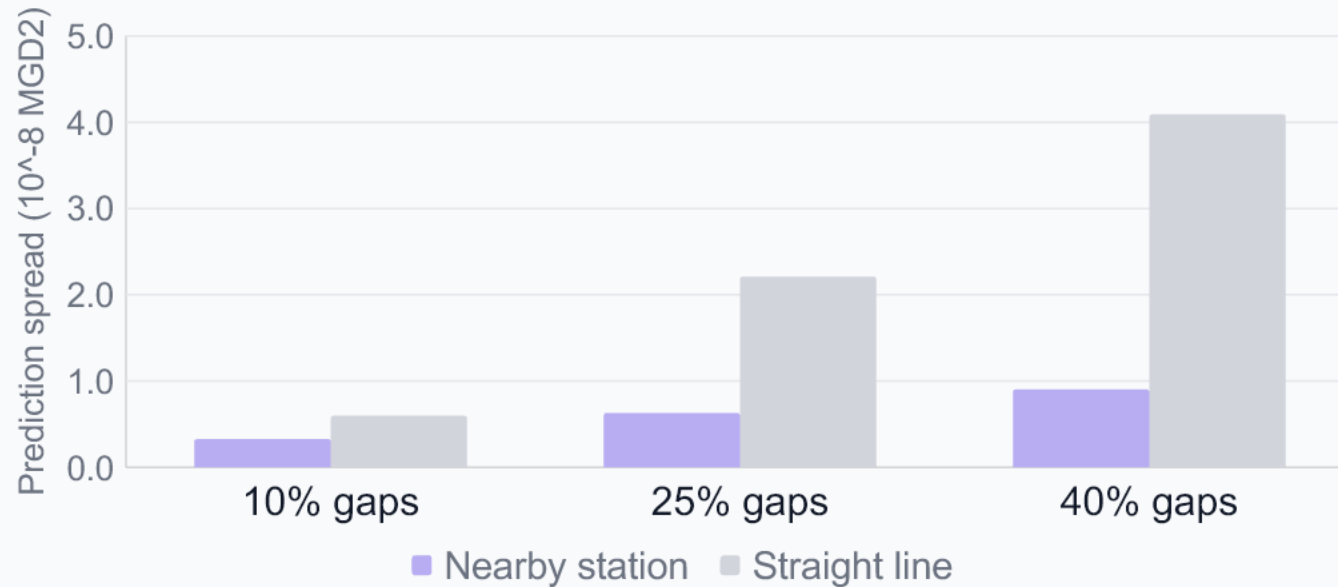
Pick using error on the middle date group. Keep RF to compare.

Next: train and test on real measurements.

Dates: train 60% / choose + set ranges 20% / test 20%

Dry air cooling uses zero water here. Its R2 fit score cannot be calculated.

EXPERIMENT E2: MISSING READINGS



100%

All main tests passed

Change stayed within ± 0.05 MGD
50 repeats on 121 drought days

FAILURE: high readings
only 60.3% within limits

Next: test bad neighbor readings and stations losing data together. The bad-reading test uses a simple formula.

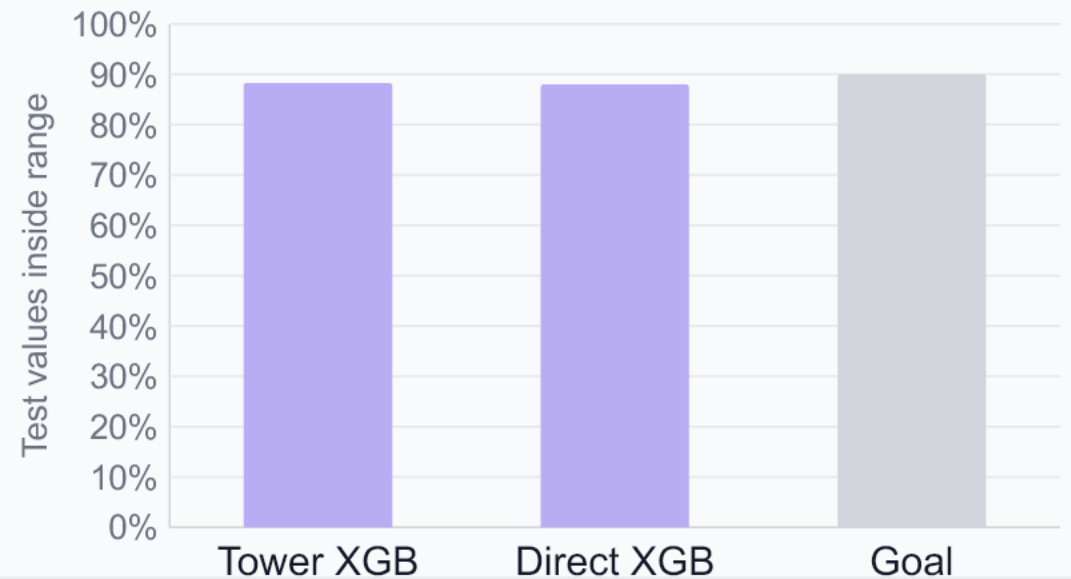
EXPERIMENT E₅: PREDICTION RANGES

88.3% / 88.0%

RELIABILITY ISSUE: below the 90% goal

Tower fit score (R^2) = 0.998

A high score cannot promise a useful range.



Improve: use separate data to choose models and set ranges.

A small test. We have not checked if the shortfall is more than chance. Real-world range tests are still needed.

DECISIONS & NEXT QUESTION

KEEP (ADOPT)

E1: lower XGBoost error. E3/E4: checks pass.
Keep the models, limits, and local matches.

CHANGE (MODIFY)

E2: bad neighbor readings. E5: below 90%.
Train on real data. Improve gaps/ranges + area totals.

DO NOT USE (REJECT)

E5: a fit score cannot promise a right answer.
Cooling use alone cannot show ecosystem health.

WAIT (DEFER)

Need real area data before testing more AI methods.

Can training on real data predict effects on water reserves and rivers in a new area?

Shaun: code + connect area data
Troy: data + source quotes
Scott: tests + results